Name:-Aditi Yedge.
Roll No:- 68
Batch:-A2
Subject:- DEVOPS

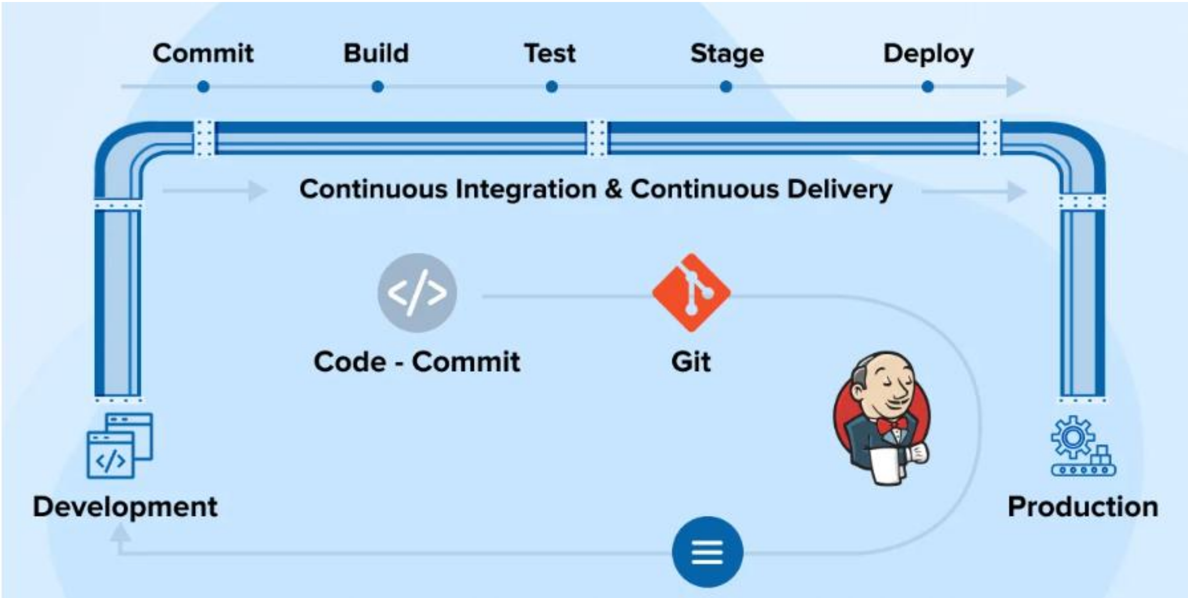
Assignment 3

Aim: Demonstrate Continuous Integration and development using Jenkins.

What is Jenkins Pipeline?

Jenkins Pipeline (or simply "Pipeline" with a capital "P") is a suite of plugins which supports implementing and integrating *continuous delivery pipelines* into Jenkins.

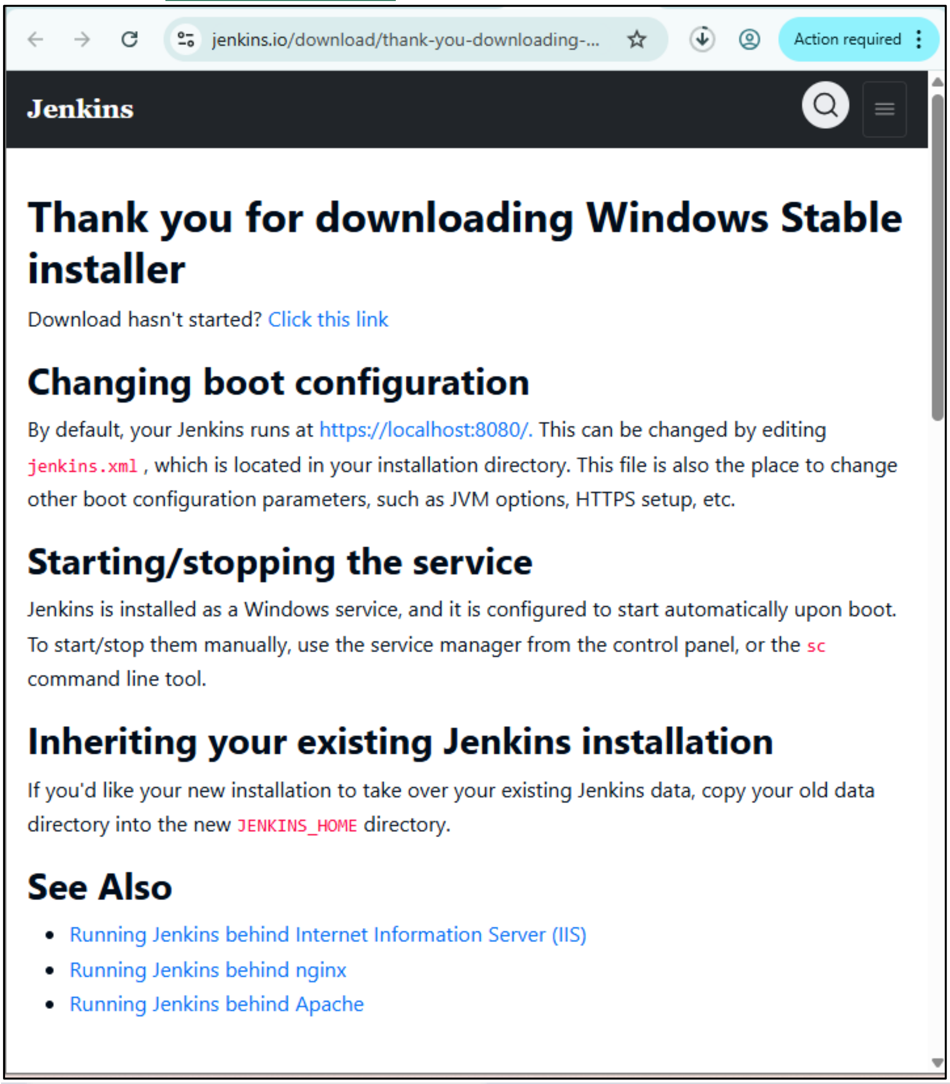
A *continuous delivery (CD) pipeline* is an automated expression of your process for getting software from version control right through to your users and customers. Every change to your software (committed in source control) goes through a complex process on its way to being released. This process involves building the software in a reliable and repeatable manner, as well as progressing the built software (called a "build") through multiple stages of testing and deployment.

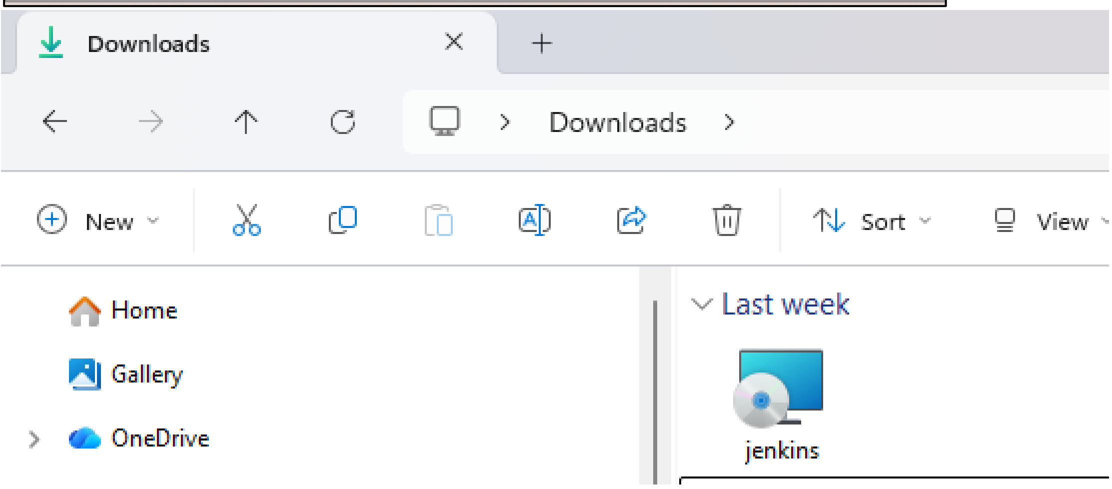


Name:-Aditi Yedge Roll No:-68 Batch:-A2 Subject:-DEVOPS

Step 1. Download the Jenkins WAR File.

Click on the Jenkins download for Windows.



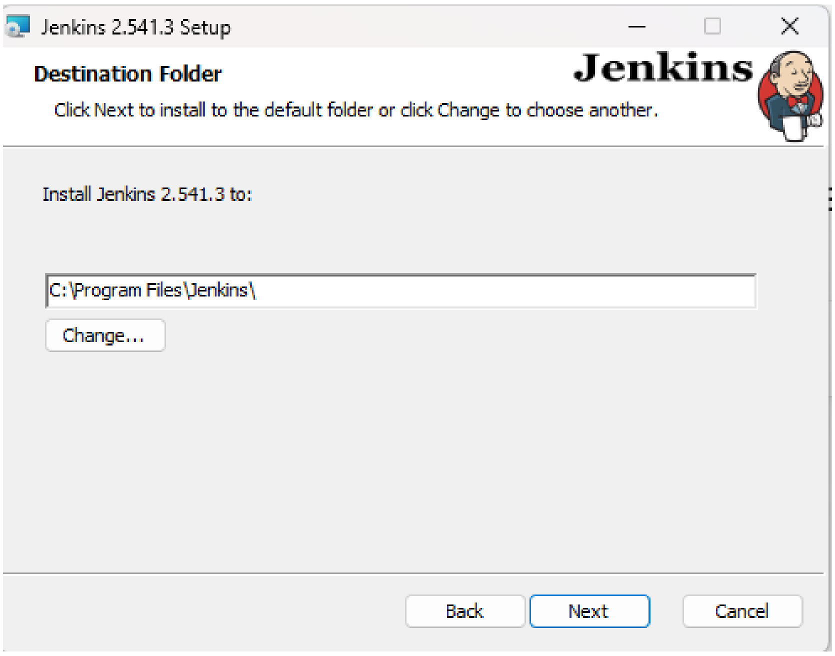


Step2: Setup wizard

When you open the Windows Installer, an **Installation Setup Wizard** appears. Click **Next** on the Setup Wizard to start the installation.

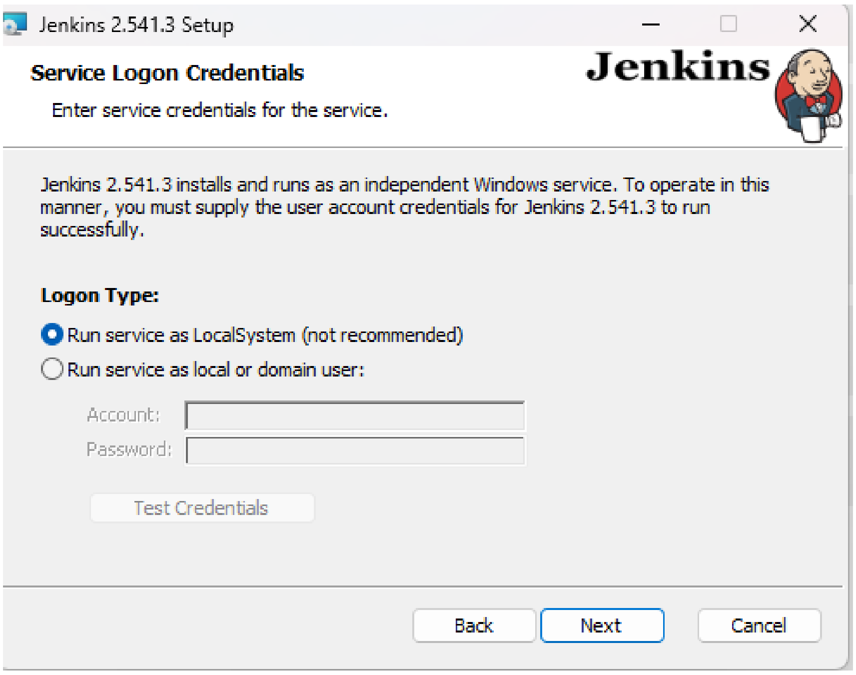
Step 2: Select destination folder

Select the destination folder to store your Jenkins Installation and click **Next** to continue.



Step 3: Service logon credentials

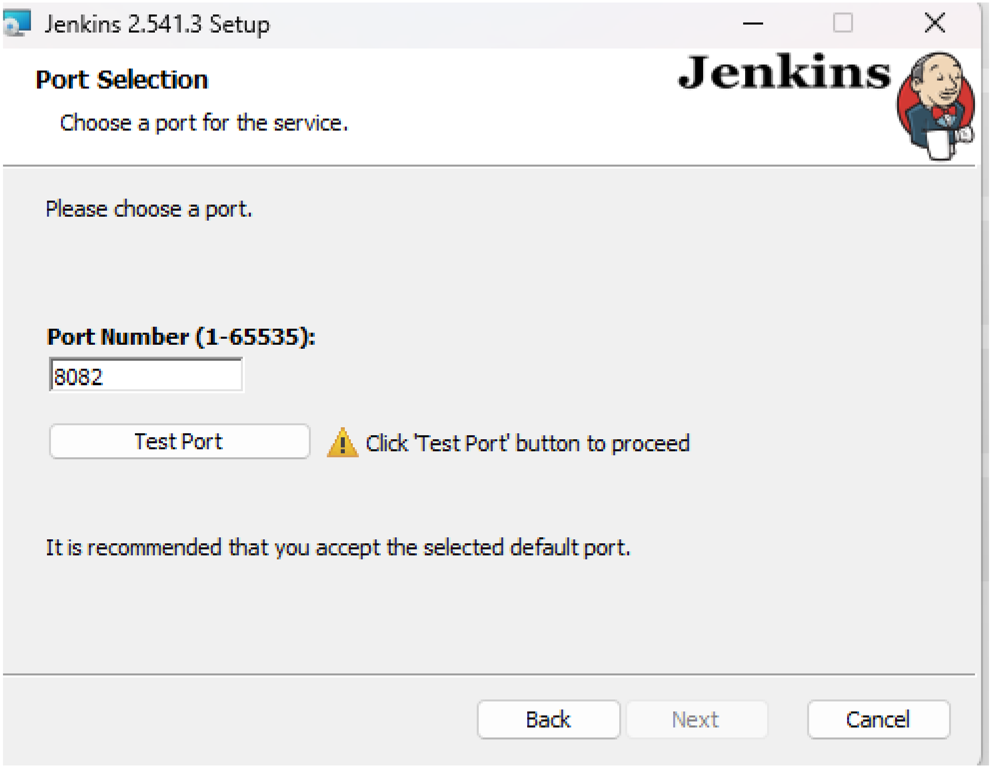
When Installing Jenkins, it is recommended to install and run Jenkins as an independent Windows service using a **local or domain user**, as it is much safer than running Jenkins using **LocalSystem(Windows equivalent of root)** which will grant Jenkins full access to your machine and services. To run Jenkins service using a **local or domain user**, specify the domain user name and password with which you want to run Jenkins,click on **Test Credentials** to test your domain credentials and click on **Next**.

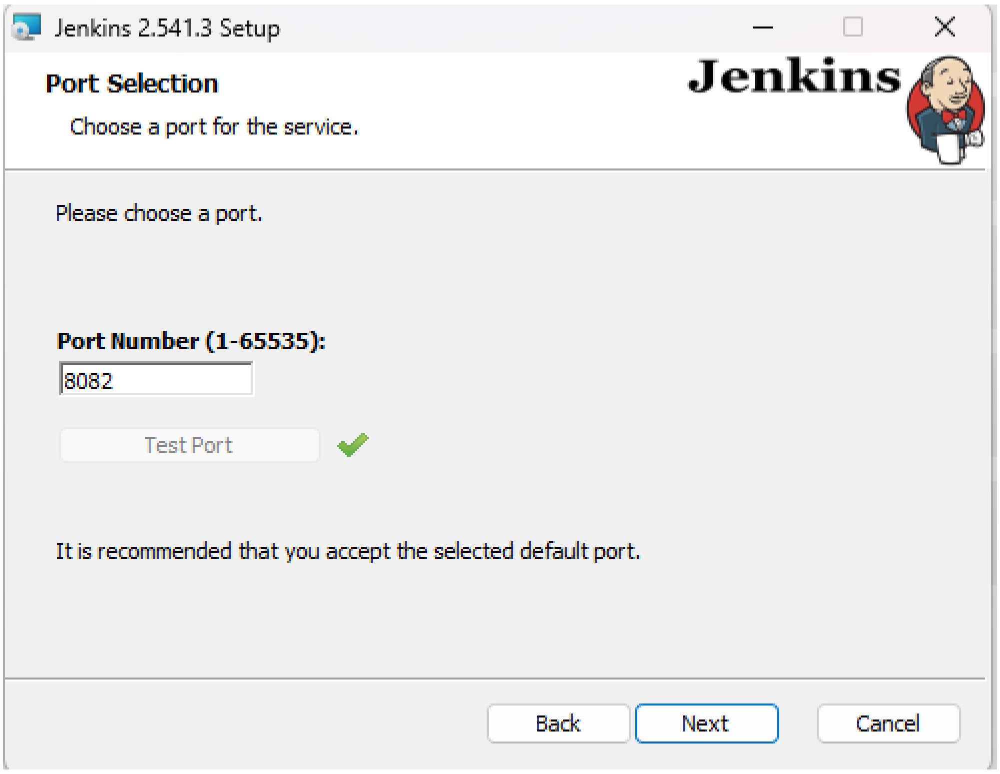


Step 4: Port selection

Specify the port on which Jenkins will run. Use the **Test Port** button to validate whether the specified port is free on your machine.

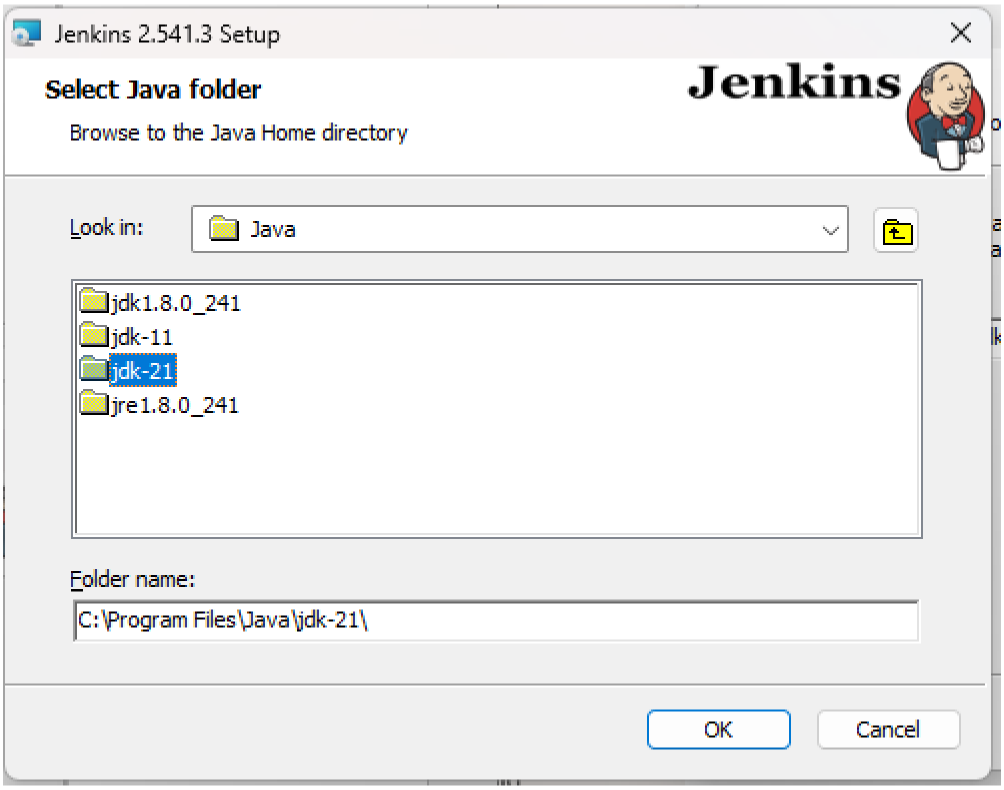
Consequently, if the port is free, it will show a green tick mark as shown below. Then, click on **Next**.

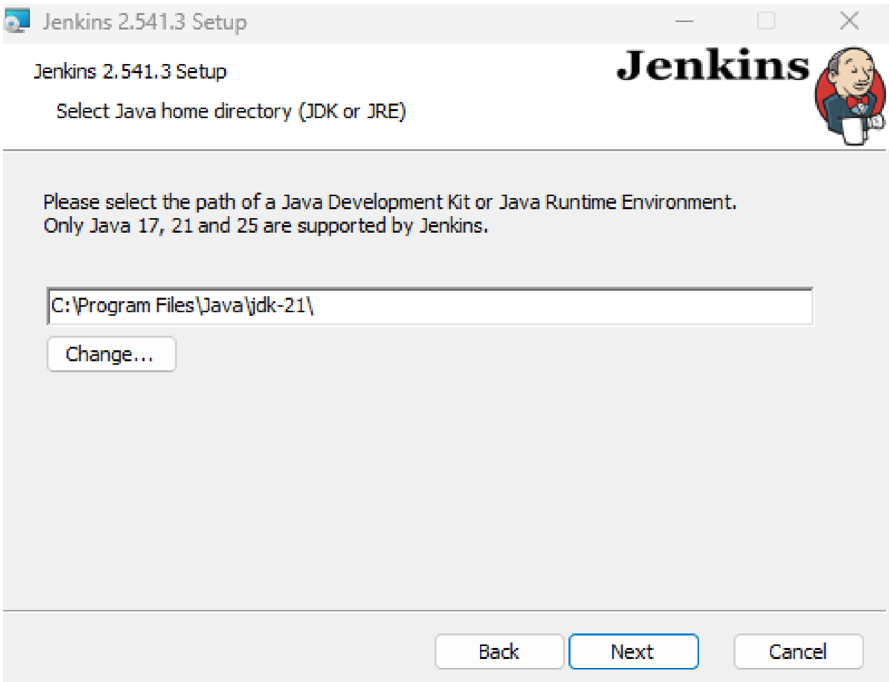

Jenkins 2.541.3 Setup
Port Selection
Choose a port for the service.
Jenkins
Please choose a port.
Port Number (1-65535):
8082
Test Port
Click 'Test Port' button to proceed
It is recommended that you accept the selected default port.
Back
Next
Cancel


Jenkins 2.541.3 Setup
Port Selection
Choose a port for the service.
Jenkins
Please choose a port.
Port Number (1-65535):
8082
Test Port
It is recommended that you accept the selected default port.
Back
Next
Cancel

Step 5: Select Java home directory

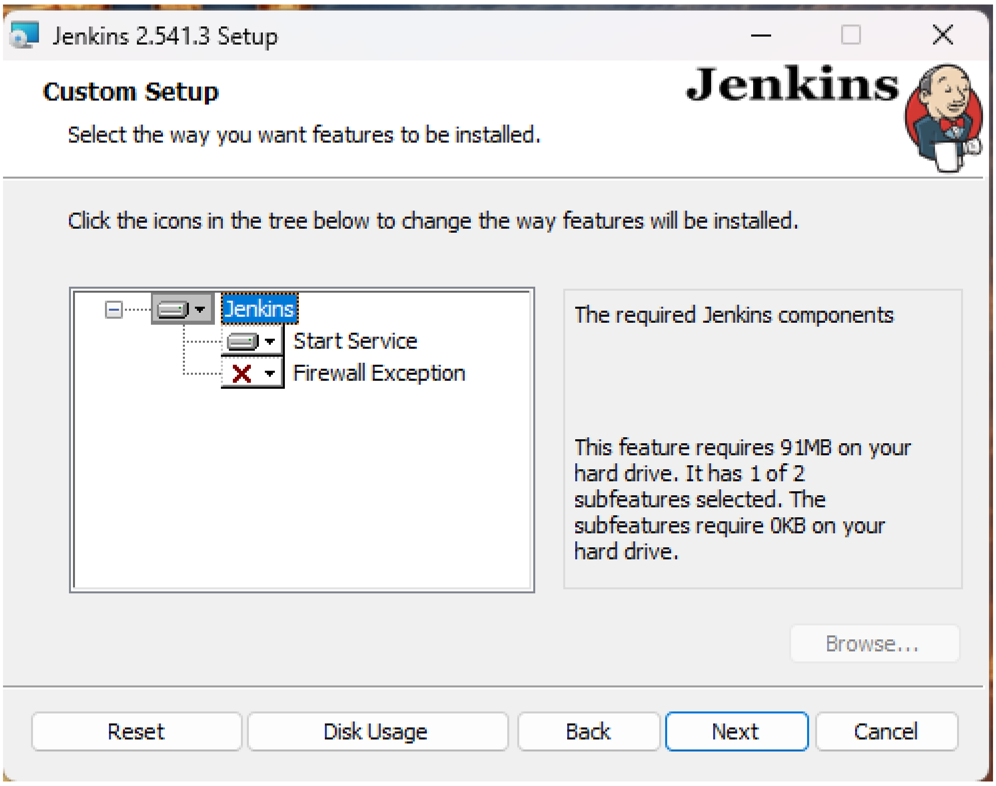
The installation process checks for Java on your machine and prefills the dialog with the Java home directory. If the needed Java version is not installed on your machine, you will be prompted to install it. Once your Java home directory has been selected, click on **Next** to continue.





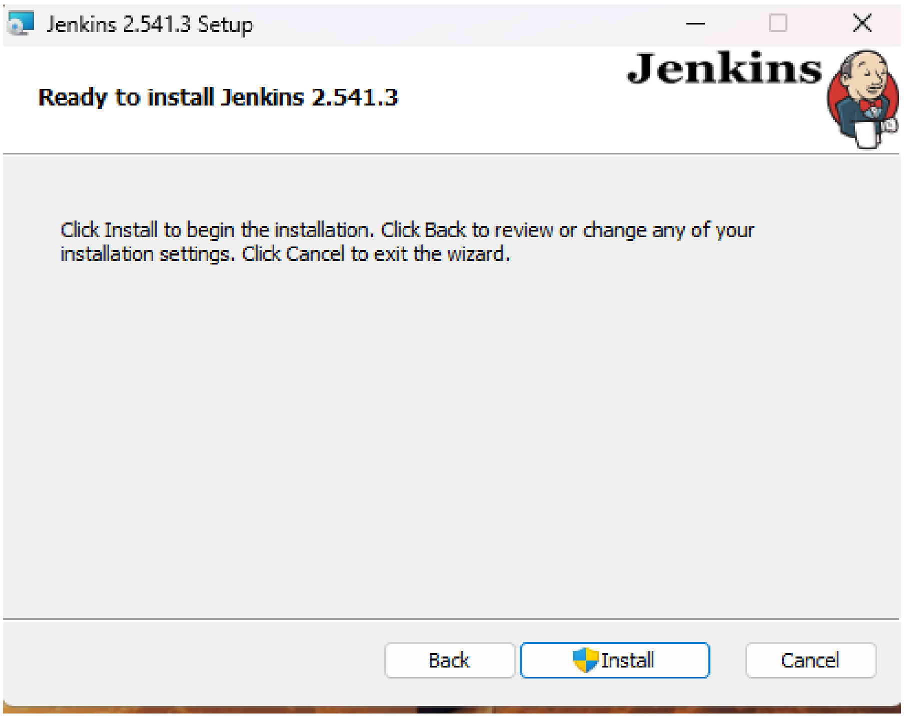
Step 6: Custom setup

Select other services that need to be installed with Jenkins and click on **Next**.

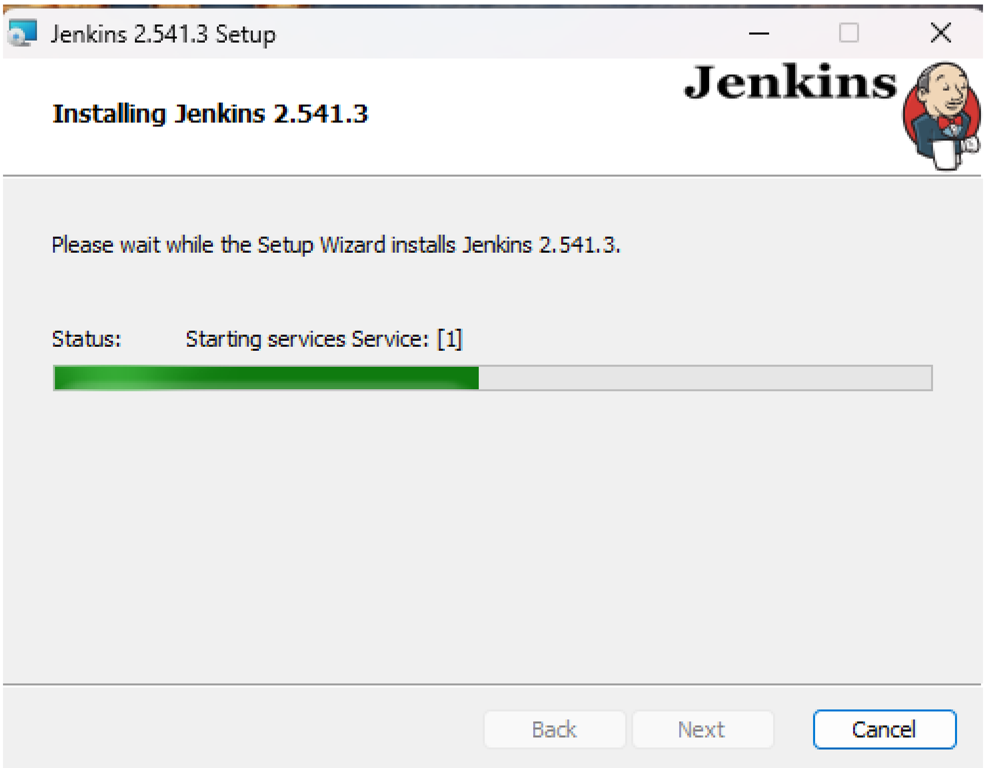


Step 7: Install Jenkins

Click on the **Install** button to start the installation of Jenkins.

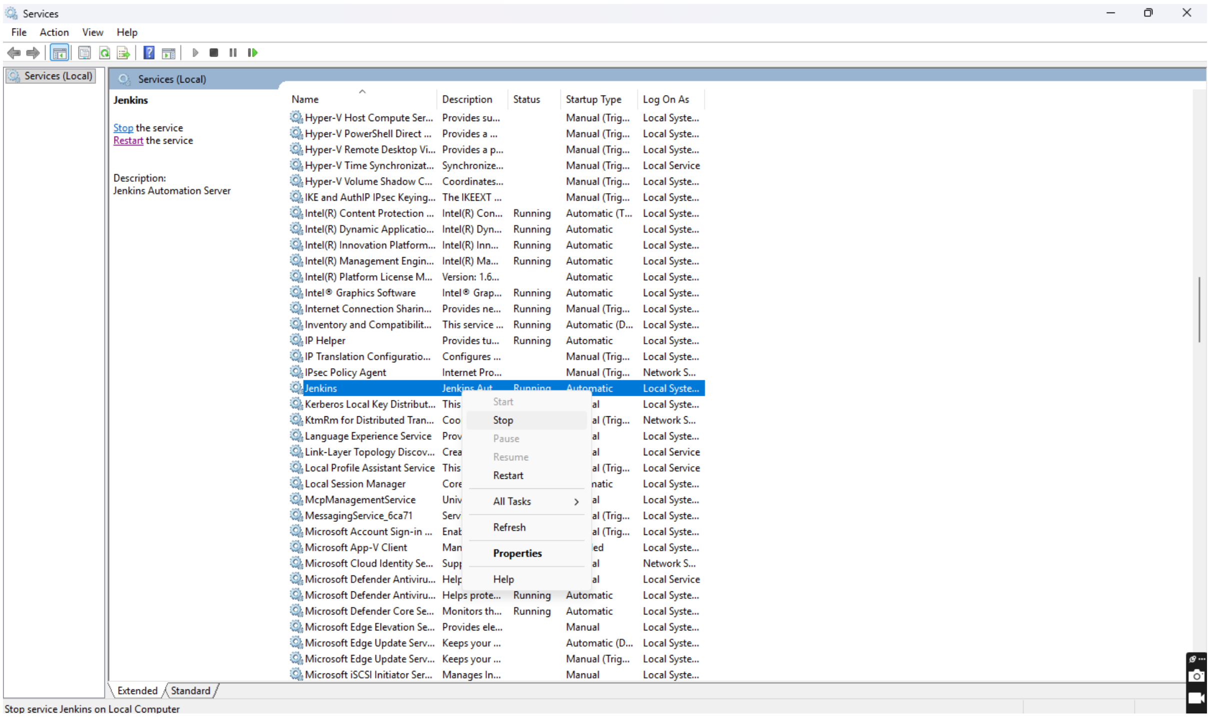


Additionally, clicking on the **Install** button will show the progress bar of installation, as shown below:



Step 8: Finish Jenkins installation

Once the installation completes, click **Finish** to complete the setup



Name:-Aditi Yedge Roll No:-68 Batch:-A2 Subject:-DEVOPS

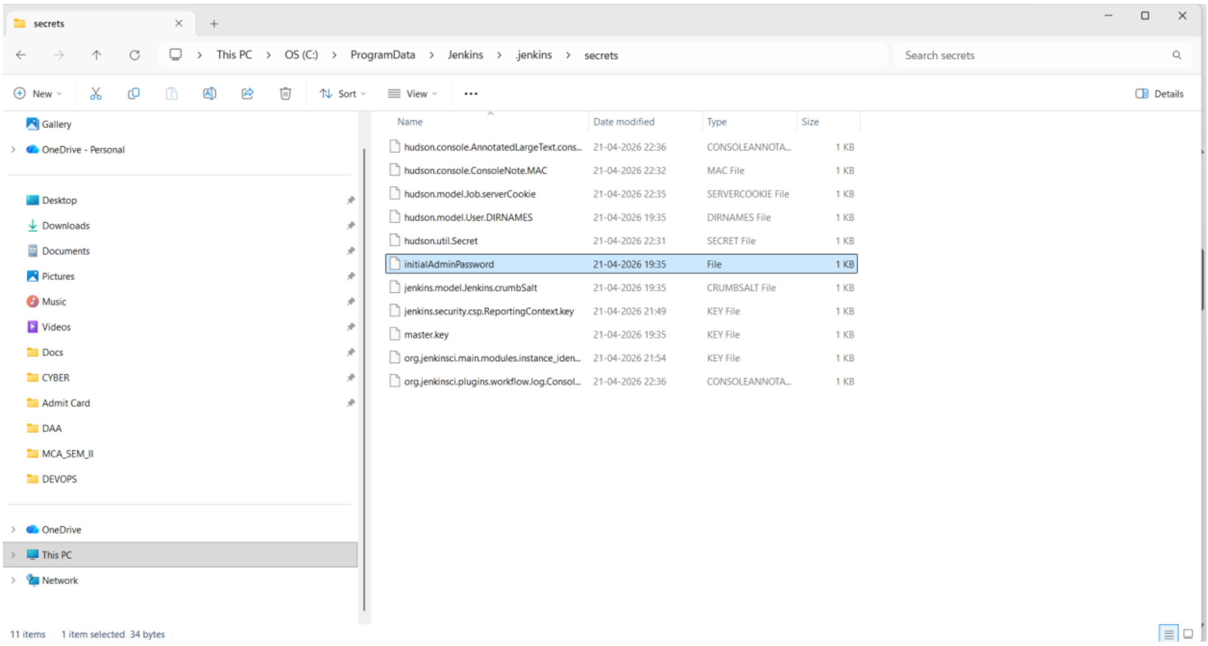
Step 9:- Sign in to jenkin

Use Username:-Admin

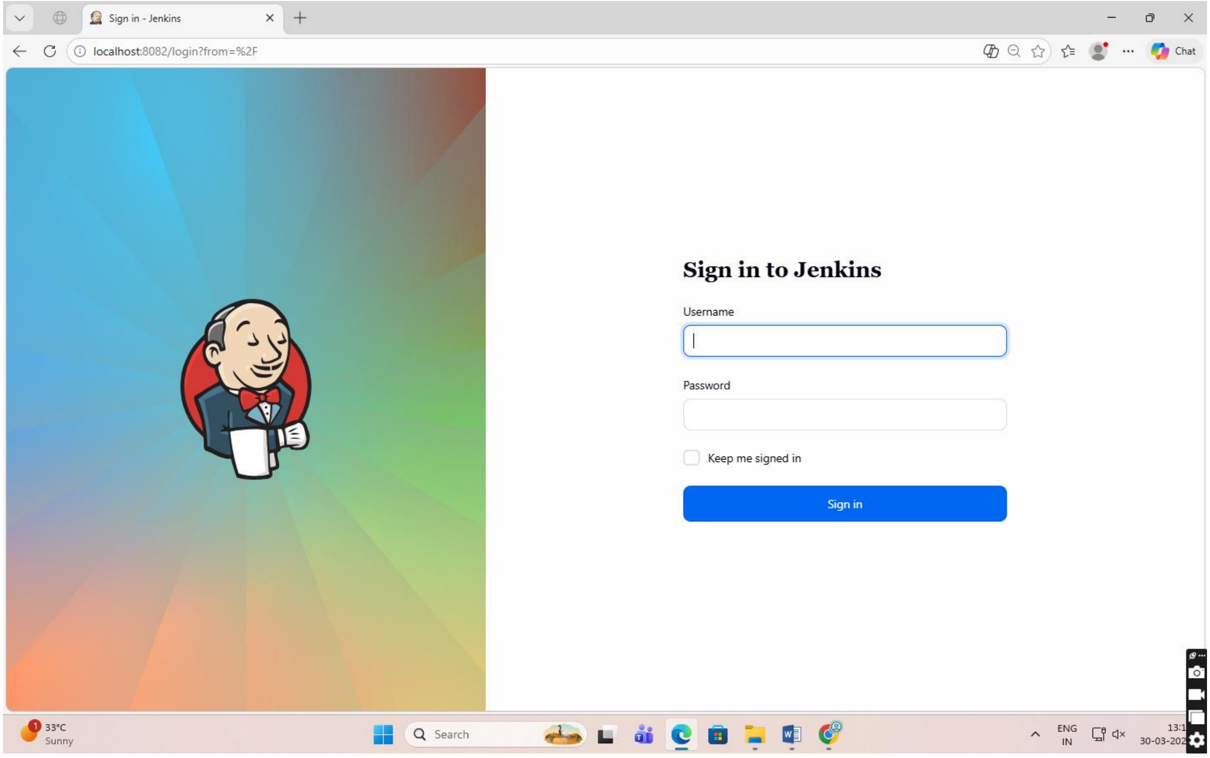
Password :- Follow Path

Program Data(S drive Hidden folder) -> Jenkins->.jenkins -> Secrets -> initialadminpassword

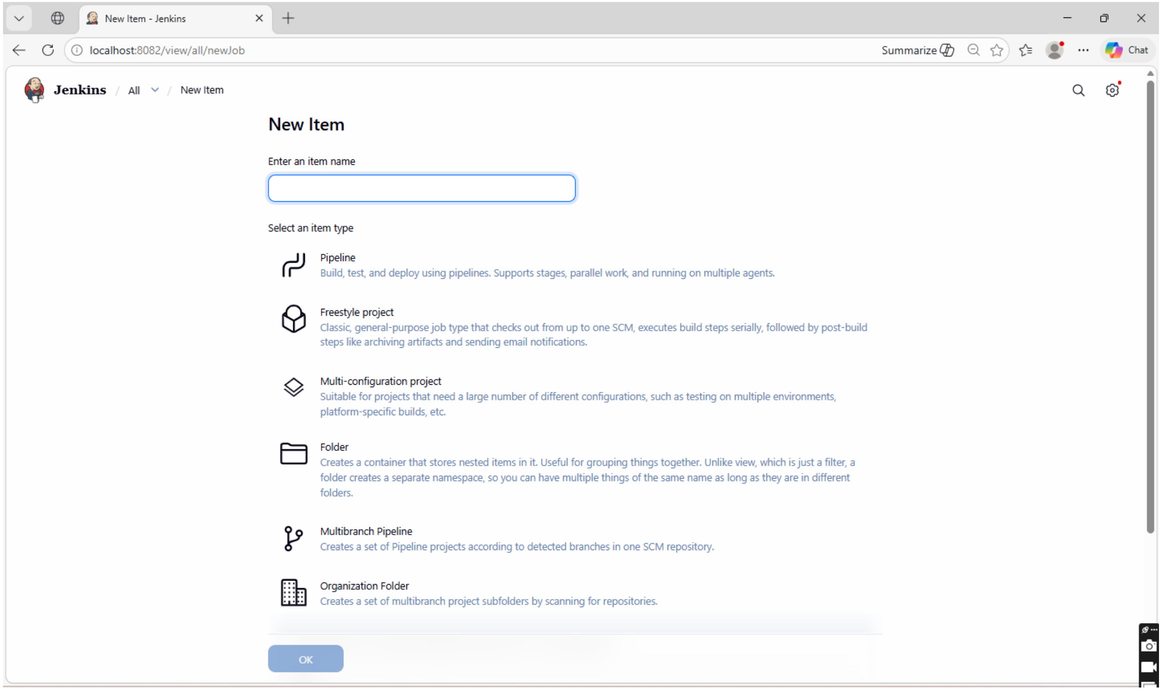
C:\ProgramData\Jenkins\.jenkins\secrets



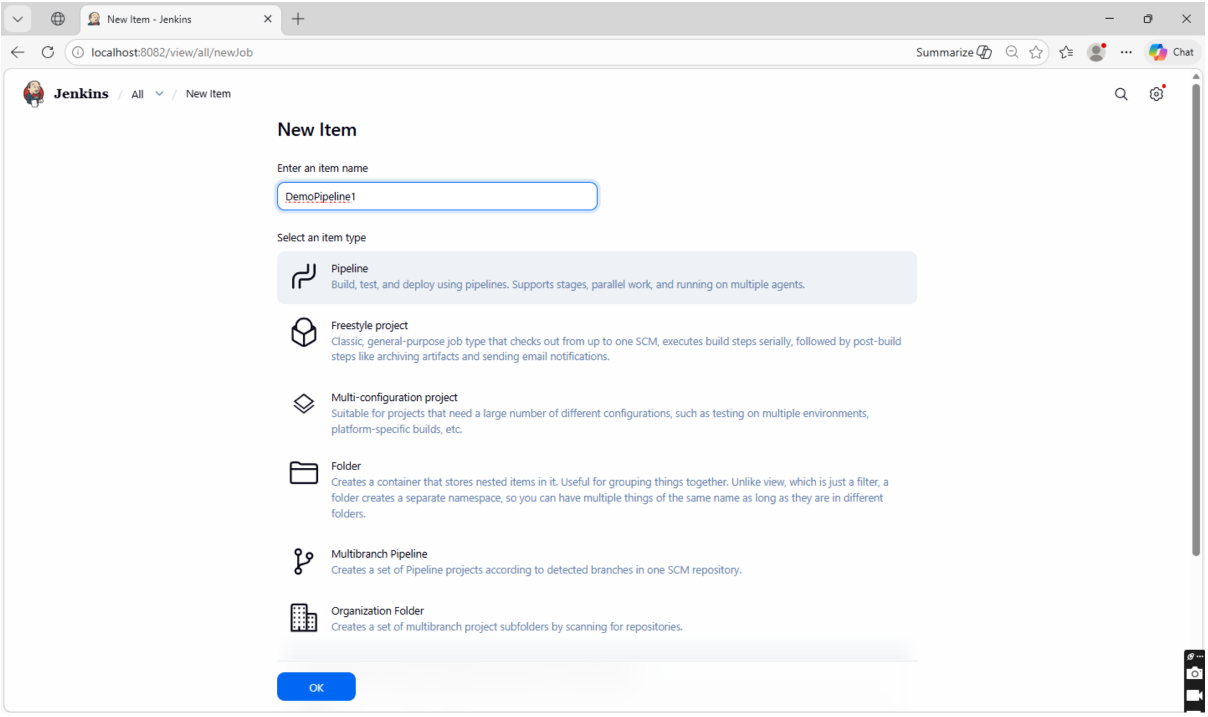
Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**

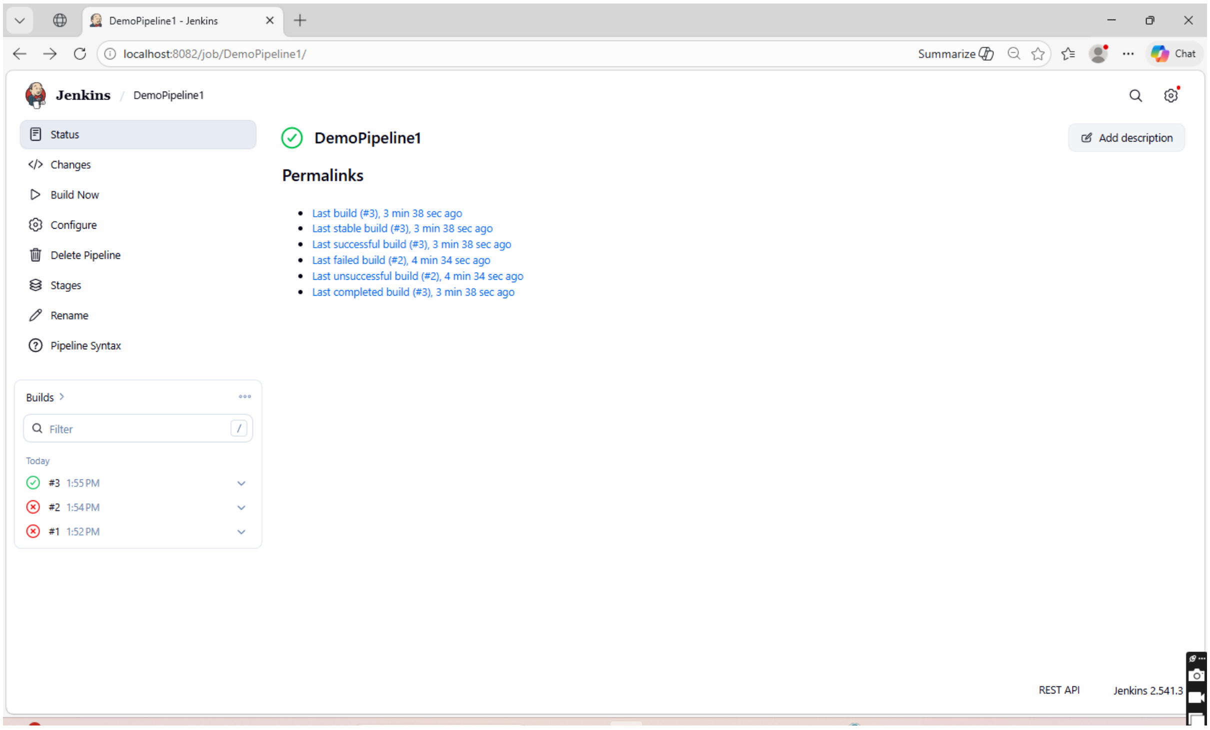


Step1:- New Item → Demo Script →Write Script→Build Now

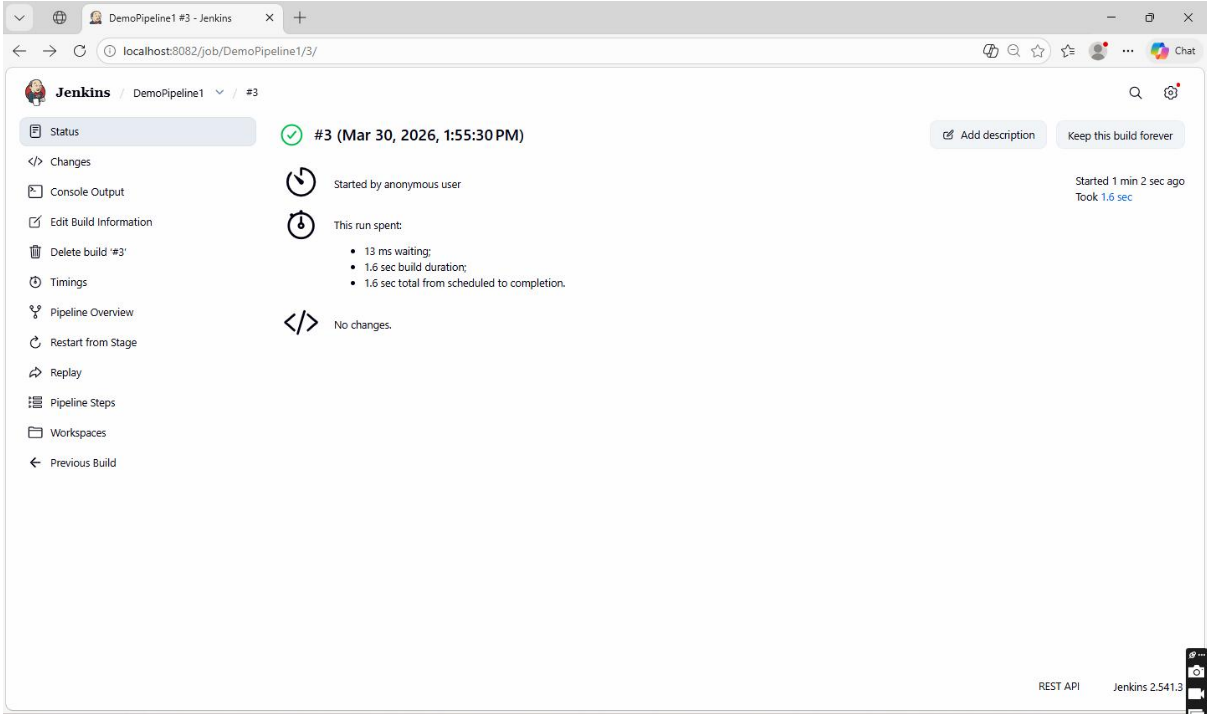


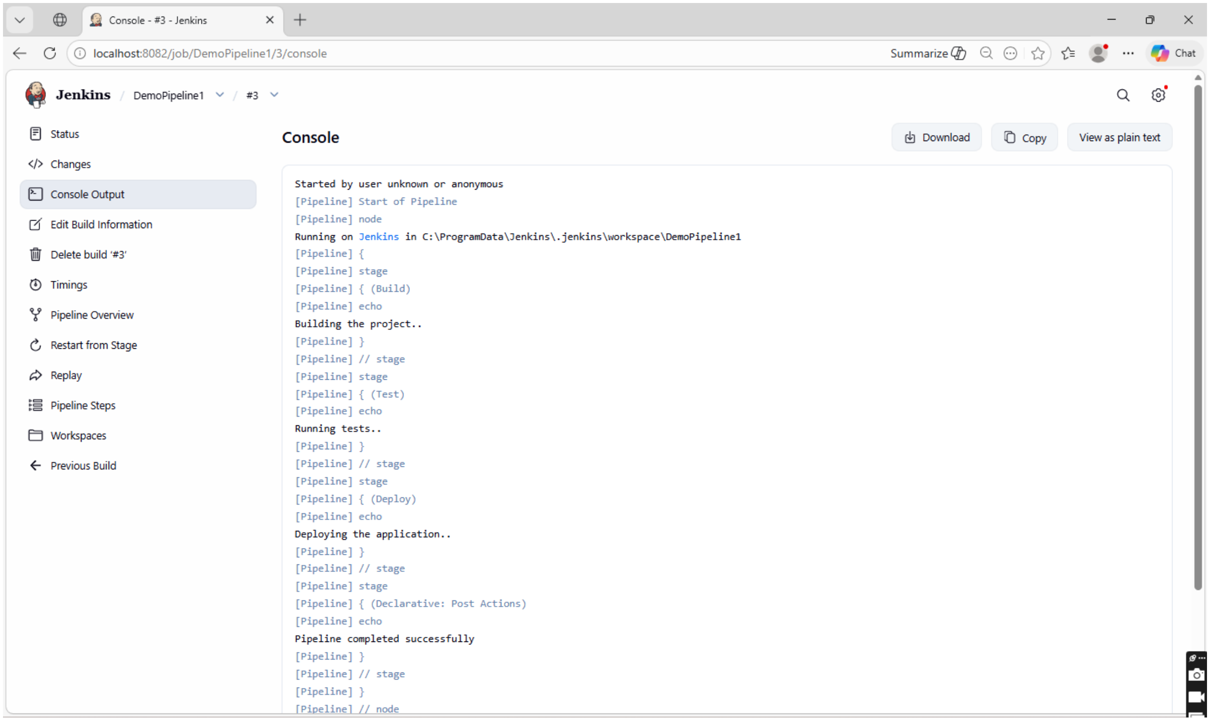
Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**





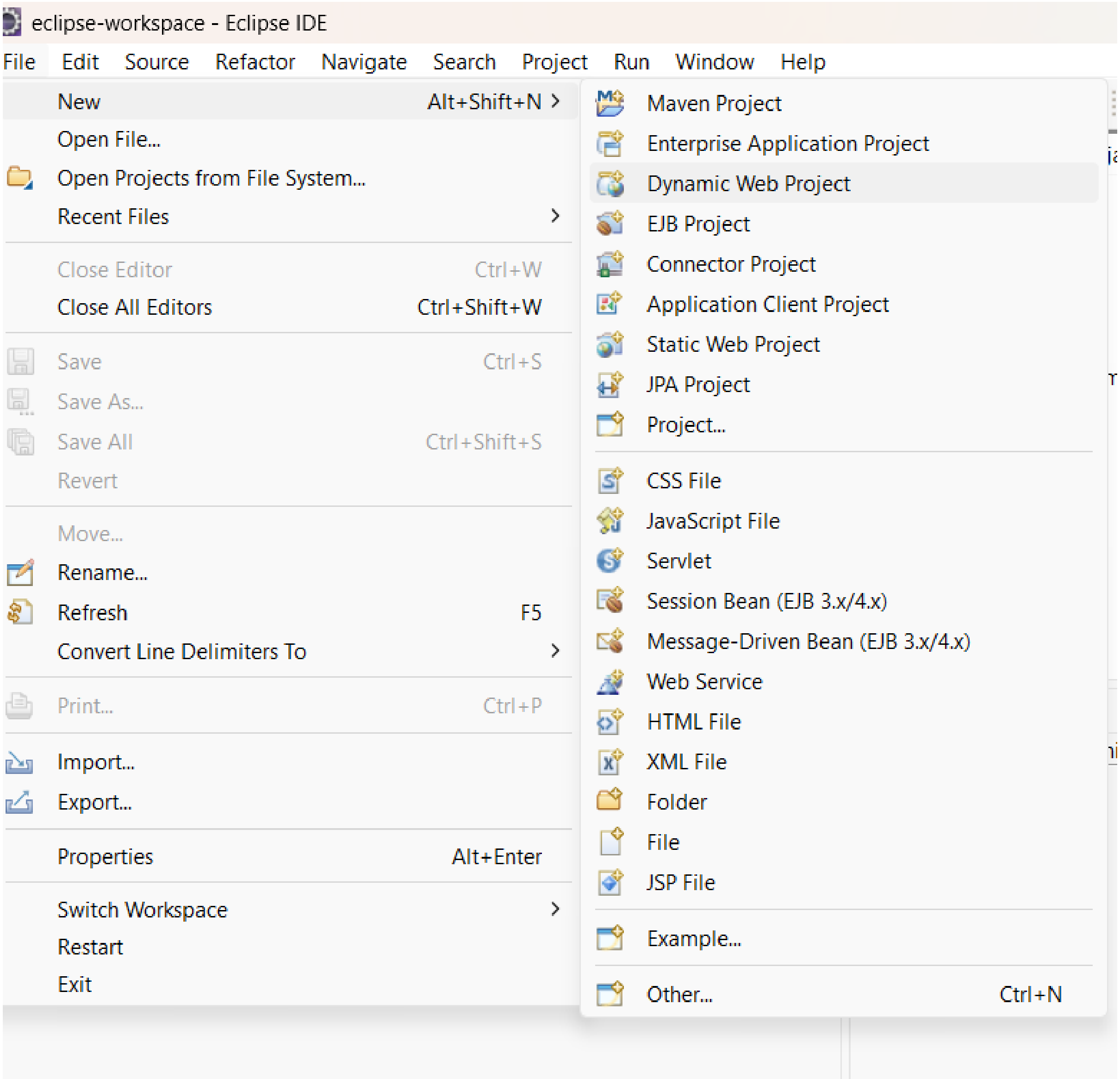
Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**



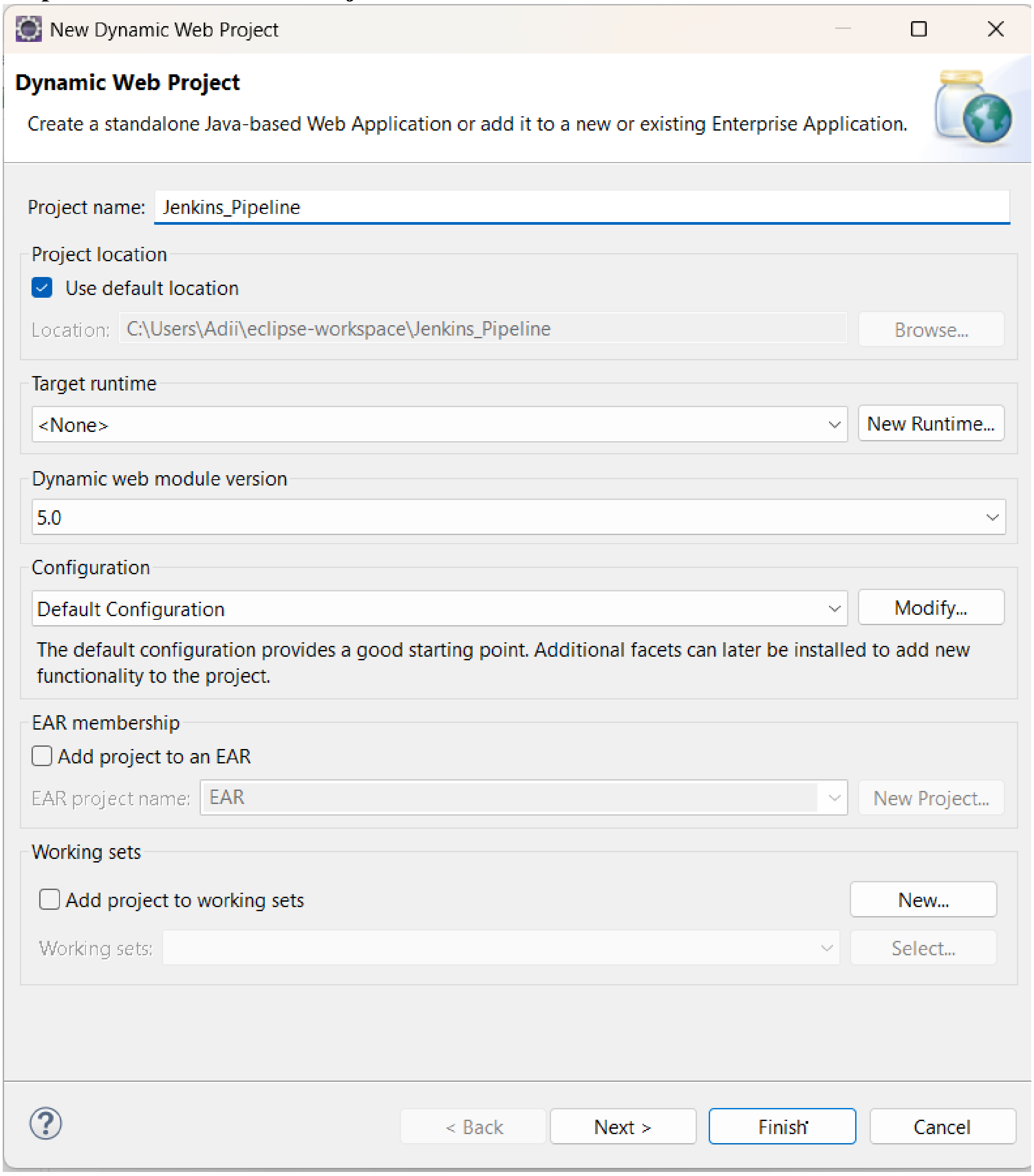


#Jenkin Pipeline Using Web Application

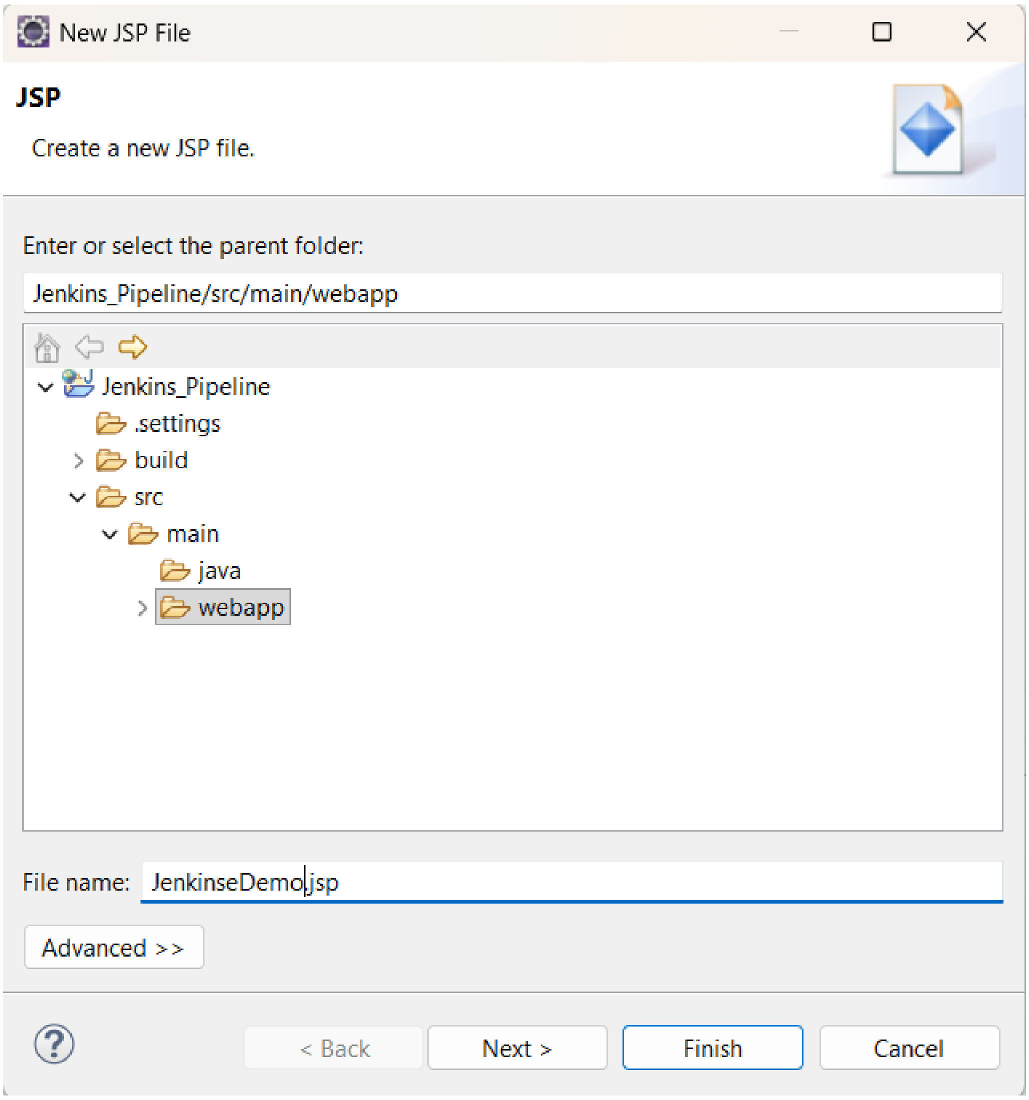
Step1:- Create Dynamic WEB Project in Eclipse.



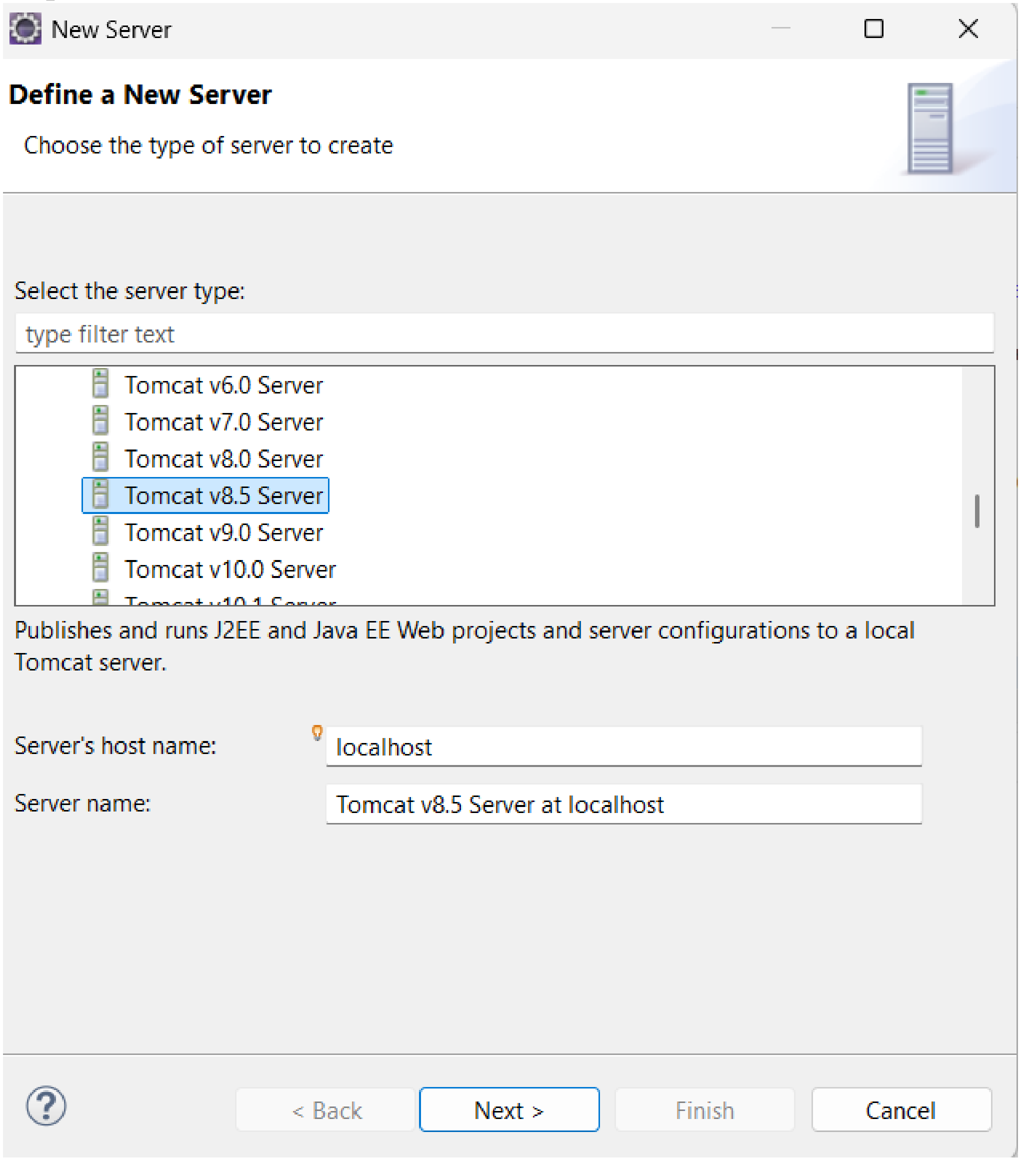
Step 2:- Give a Name to Project



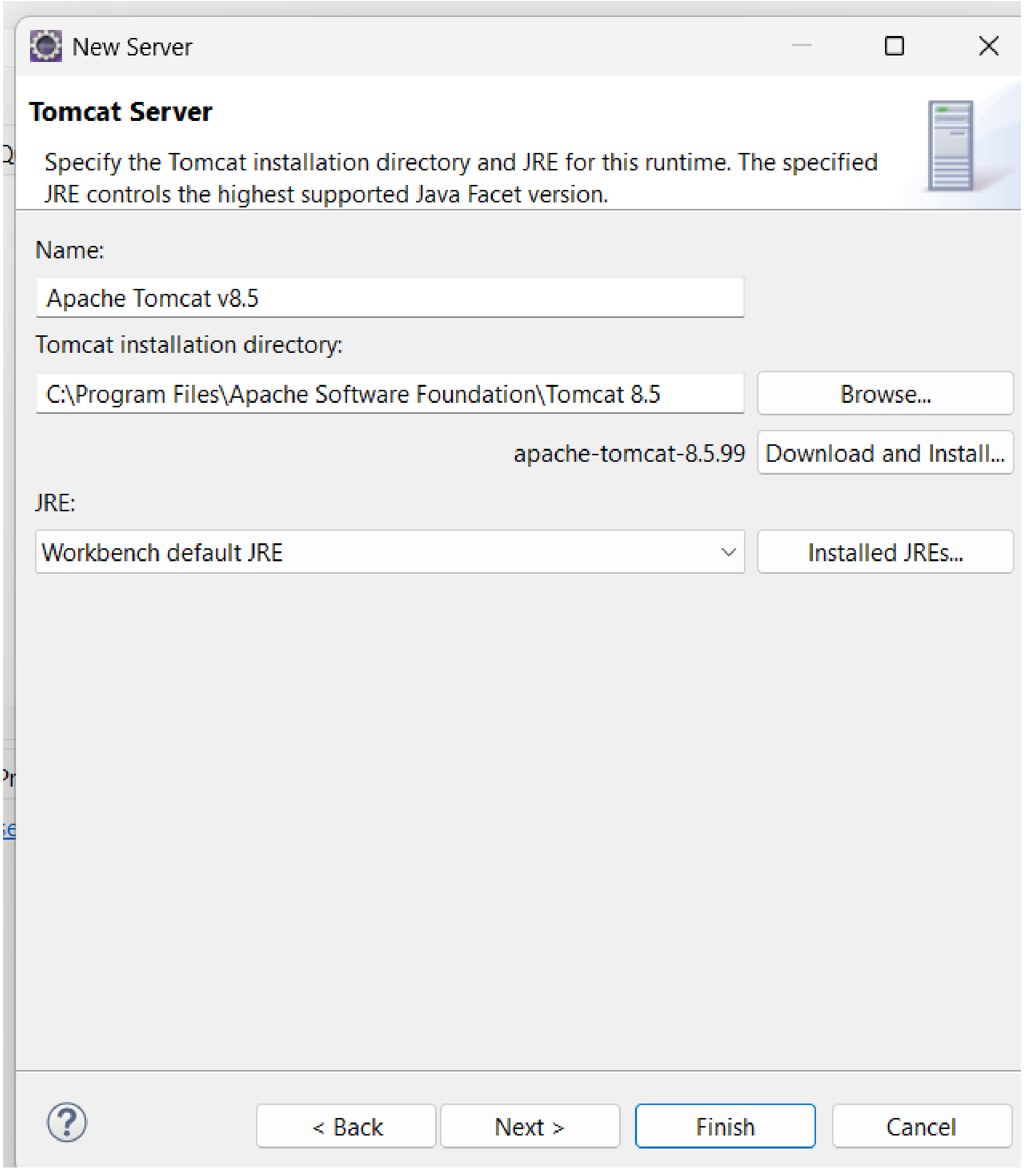
Name:-Aditi Yedge Roll No:-68 Batch:-A2 Subject:-DEVOPS



Step 3:- Select Server.



Name:-Aditi Yedge Roll No:-68 Batch:-A2 Subject:-DEVOPS

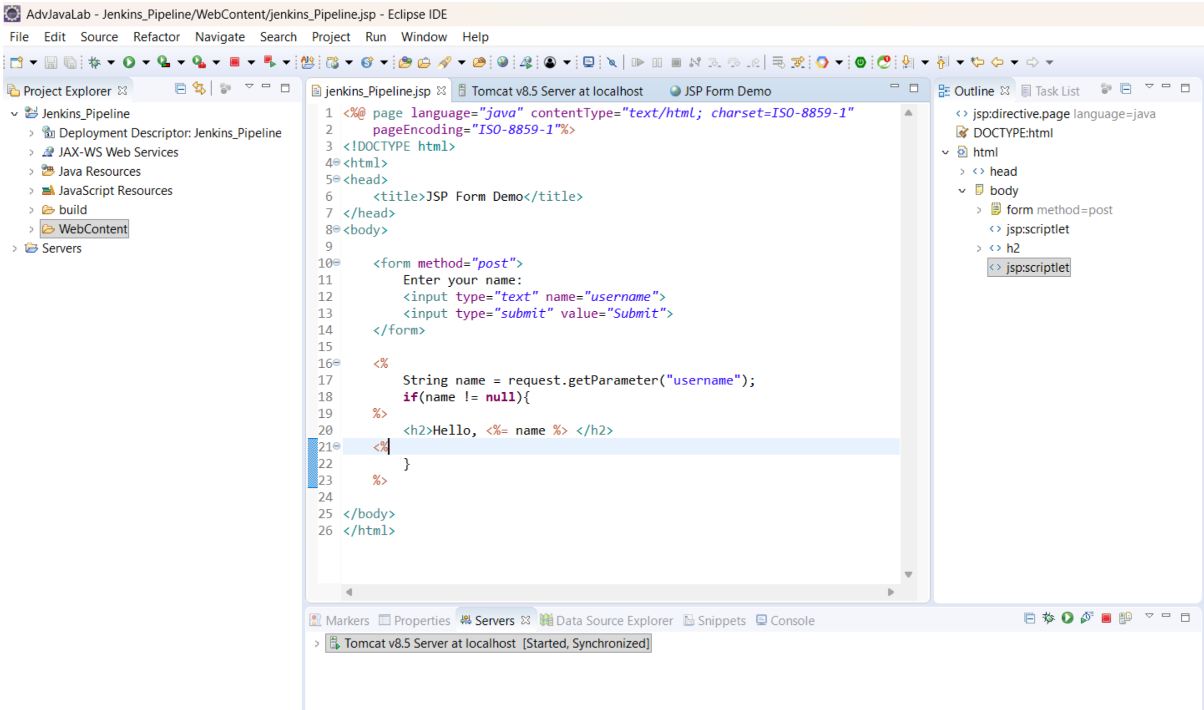


Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**

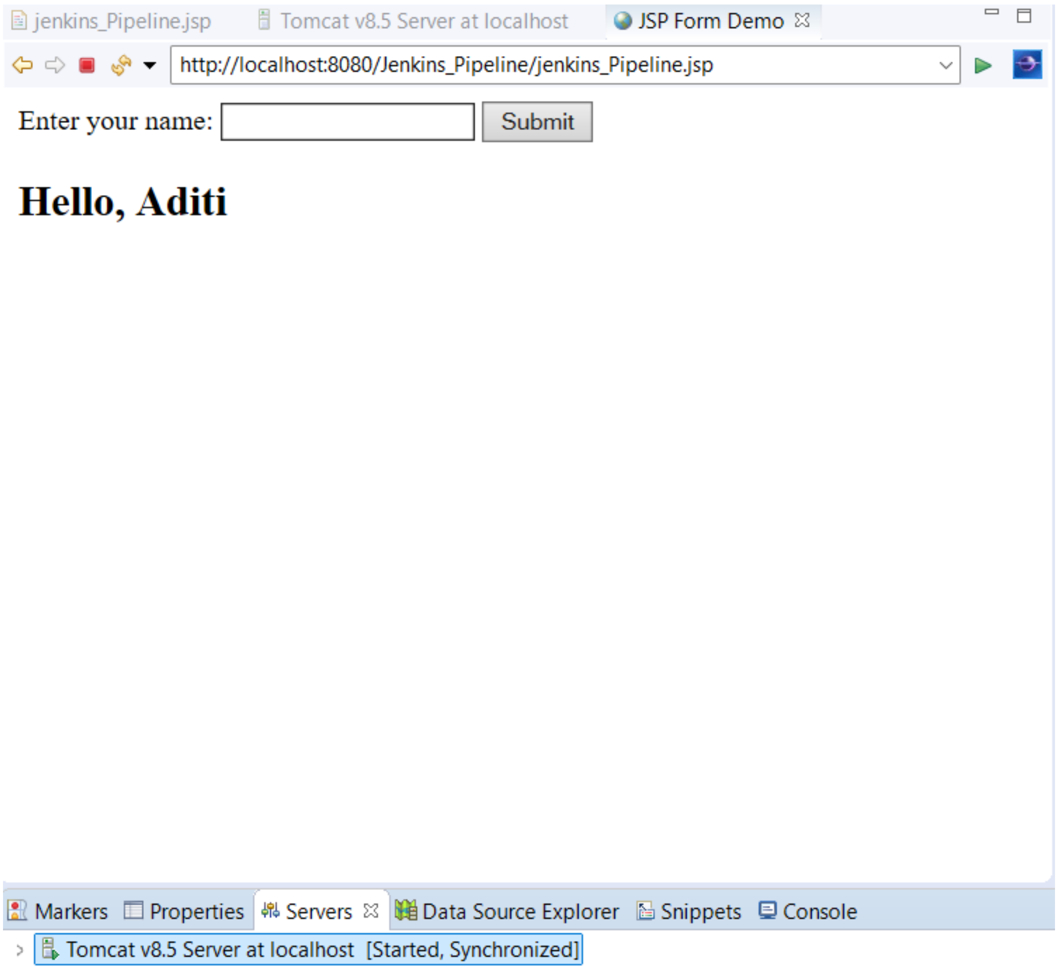
SourceCode:-

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
   pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
   <title>JSP Form Demo</title>
</head>
<body>

   <form method="post">
      Enter your name:
      <input type="text" name="username">
      <input type="submit" value="Submit">
   </form>

   <%
      String name = request.getParameter("username");
      if(name != null){
   %>
      <h2>Hello, <%= name %> </h2>
   <%
      }
   %>

</body>
</html>
```

Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**

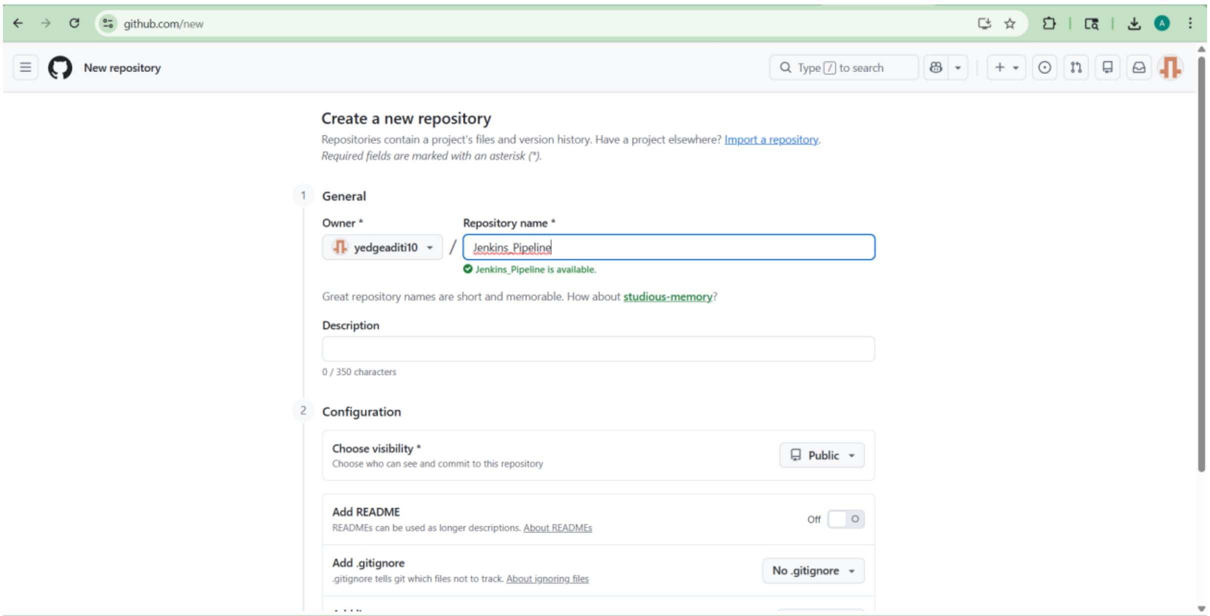


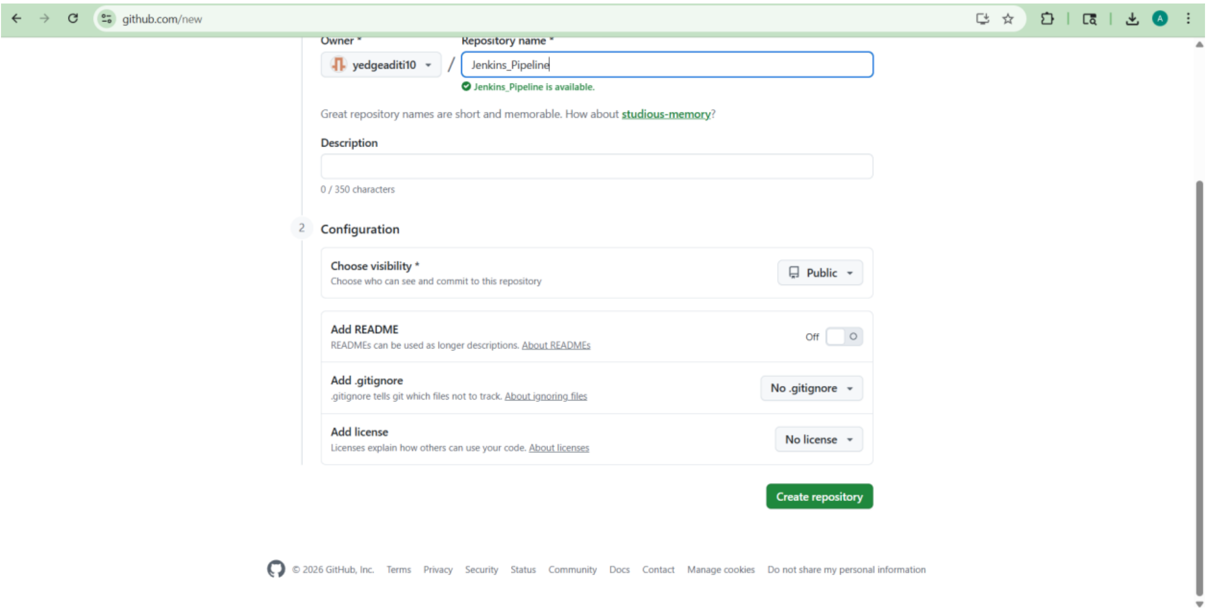
OUTPUT:-



Step 4:- Log in to GitHub Account.

1.Cretae A New Repository.



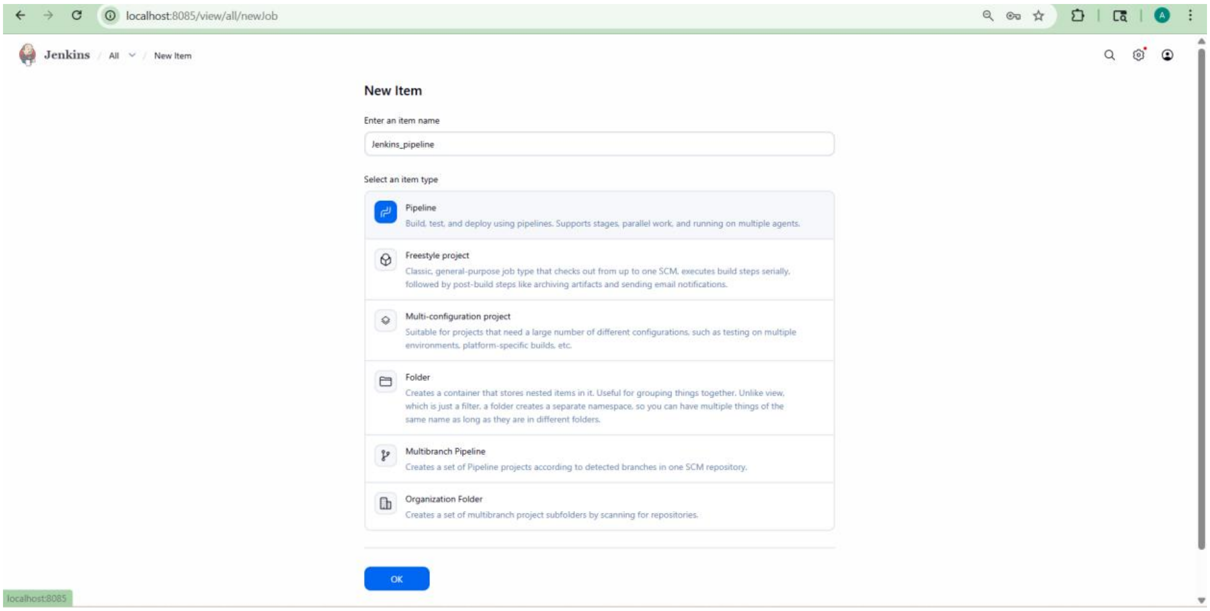


Step 5:- Open Git Bash & follow the commands.

Name:-Aditi Yedge Roll No:-68 Batch:-A2 Subject:-DEVOPS

```
MINGW64:/c/Users/Adii/Desktop/DEVOPS/Jenkins_Pipeline

Adii@DESKTOP-C60SFL6 MINGW64 ~ (master)
$ cd "Desktop\DEVOPS\Jenkins_Pipeline"

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git init
Reinitialized existing Git repository in C:/Users/Adii/Desktop/DEVOPS/Jenkins_Pi
peline/.git/

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git add .

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git remote add origin https://github.com/yedgeaditi10/Jenkins_Pipeline.git
error: remote origin already exists.

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git branch -M main

Adii@DESKTOP-C60SFL6 MINGW64 ~/Desktop/DEVOPS/Jenkins_Pipeline (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 8 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (16/16), 2.60 KiB | 296.00 KiB/s, done.
Total 16 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/yedgeaditi10/Jenkins_Pipeline.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

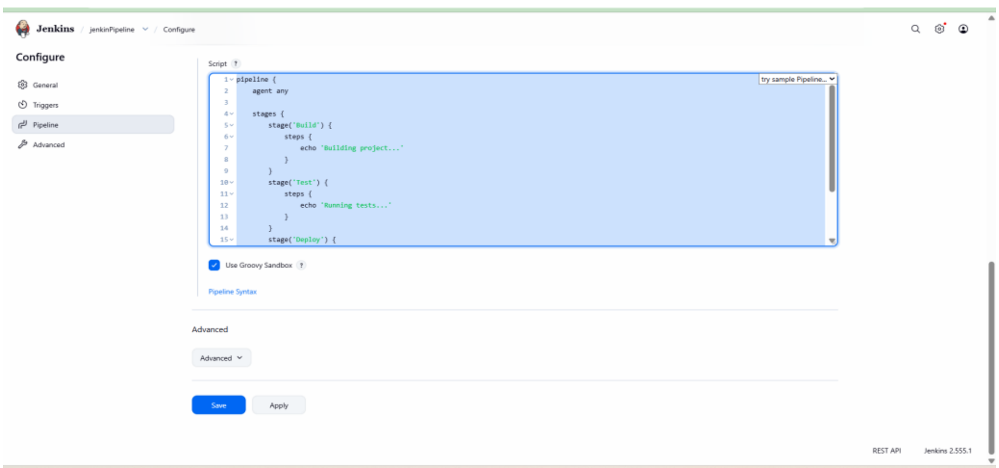
Step 6:- Create a New Item And Select Pipeline



Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**

Write Script

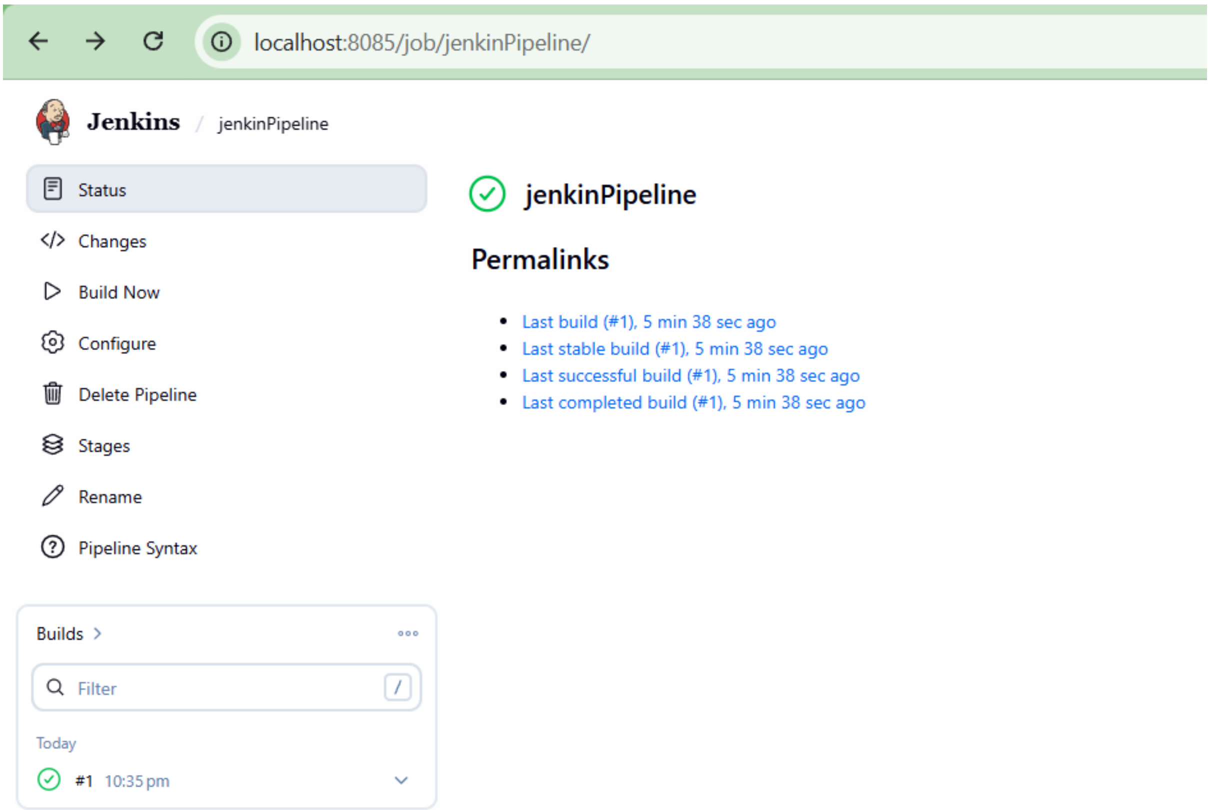
```
pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                echo 'Building project...'
            }
        }
        stage('Test') {
            steps {
                echo 'Running tests...'
            }
        }
        stage('Deploy') {
            steps {
                echo 'Deploying application...'
            }
        }
    }
```

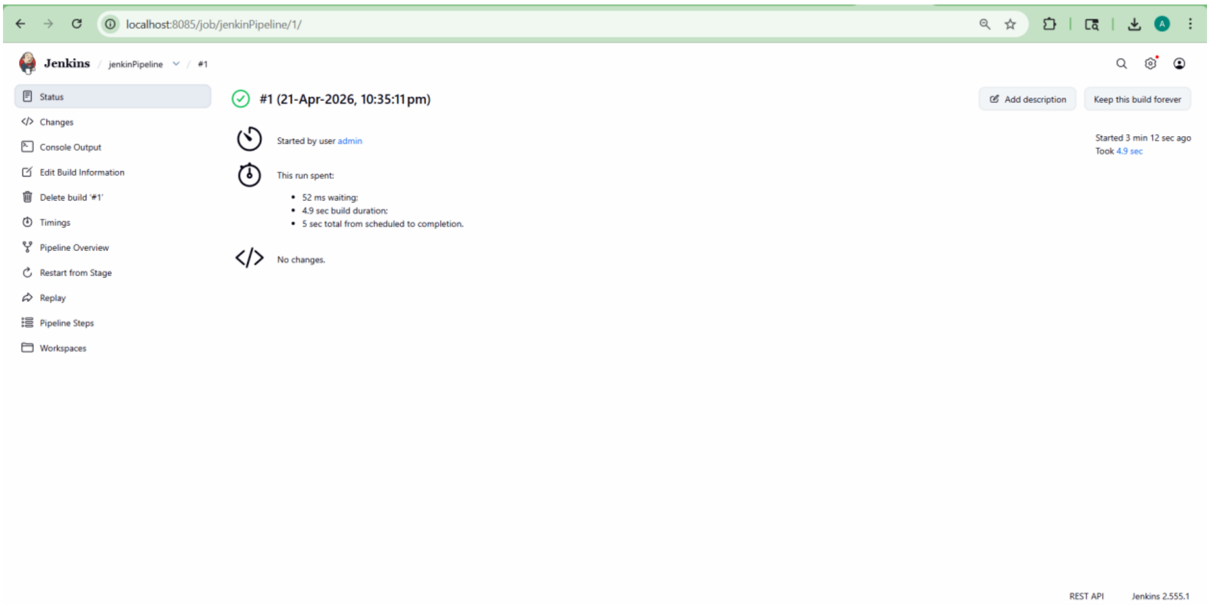



```
}
```

Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**

OUTPUT:-





Name:-Aditi Yedge **Roll No:-68** **Batch:-A2** **Subject:-DEVOPS**